

It is important that Step 2 is only done after Step 1, since device names can change from installation to installation and session to session.

 **Note:** Although removing junk files while running a live CD is fairly simple, you'll have to boot the to-be-copied OS itself to remove junk applications, unless you're familiar with `*chroot*`.

Finding the device names

In GNU/Linux, disks and partitions are usually mapped into the device files inside `/dev`.

`/dev/sdX` represents a SATA mass storage device, where X can be a, b, c, etc. A mass storage device can be anything – a hard disk, SSD or a USB Flash drive, while optical disks are usually mapped to `/dev/sr0`. When referring to a single partition, `/dev/sdXY` is used, where Y is a number. This means the first partition in your first hard disk is probably `/dev/sda1`.

Guessing is okay if you have a passion for system destruction and data removal. Otherwise, be extremely careful, because naming can vary based on different factors.

There are multiple ways to find the device names. You can use some graphical utility like GNOME Disks or *gparted*. When it comes to the console, I'd recommend a combination of *df* and *blkid* because you won't make any accidental changes that way.

df -h will print the device names, used and unused space, and the path where each one is currently mounted – all of which are sufficient to identify the name of the disk or partition that you are considering. *blkid* `DEVICE-NAME` will print the UUID and label (if set) of the partition so that you can verify.

Removing junk files and programs

Unless you are considering a bit-by-bit copy of the disk or partition, it is better to remove unwanted programs and junk

files before copying so that you can save time and protect the target disk from too much wear. However, *do not* do this if your source disk is ageing. In that case, all you want is to get the contents copied first. (If the condition of the disk is worse, it's better not to let it spin at all; seek professional service.)

Trash is an obvious collection of junk files in a desktop distro. But apart from that, there are many hidden files (the ones starting with a dot, which you can list using *Ctrl + H* in a graphical browser or using *ls -a* in a command shell) in your `/home` file. Some might be holding really important data like your browser bookmarks and security keys, but the ones like *.thumbnails* and select children of *.cache* are safe to remove. You can also make use of file decluttering tools, if you like.

There can be system-level removable data too. For example, in Debian-based distros, *sudo apt-get clean* can remove hundreds of megabytes of package management cache.

System migration is a good opportunity to remove junk applications, too. You can use commands like *apt-get purge PACKAGES* so that the configuration files are also removed. If *apt* is your package manager, another useful command is *sudo apt-get autoremove*, which will remove packages that were installed automatically and are no longer required..

Copying the files

The basic idea is to copy the files in the root partition of your existing OS to an empty partition (usually on another disk). This involves:

1. Creating a new partition of sufficient size on the destination disk (or choosing an existing one).
2. Formatting it with a compatible file system (usually *ext4* for modern GNU/Linux distros).
3. Copying all the files preserving the file metadata.

You can use console utilities like *fdisk*, *gdisk* and *mkfs.ext4* to perform the partitioning and formatting. If you are not sure, it's safer to use a graphical utility like *gparted*.

Unlike the disk imaging method, both source and destination partitions can be mounted before copying (in fact, that's necessary). But you have to make sure that no other write activity is going on. That's why copying the root of a running GNU/Linux system is not recommended. Boot into a live distro, mount the root of the to-be-copied OS to some safe place like `/mnt`, and start copying.

Copy as the root user so that there will be no permission issues (you can use *sudo*). You have to preserve file metadata also. Assuming the old root is mounted to `/mnt`, the command can be something like this:

```
sudo cp -a /mnt/* /media/liveuser/
NEWPARTITION/
```

You can use the *rsync* utility also. Both will accept the *-a* option, which preserves the metadata.

Updating *fstab*

`/etc/fstab` is the file that maps partitions to their automatic mount points. Once the copying is complete, you can edit the *fstab* file in the new partition, to replace the ID (usually a long hexadecimal value) of the old root partition with the new one. Yes, you can use *blkid* to know the ID.

You can also consider this an opportunity to bring your `/home` into or out of the root partition, which is easily done by adding or removing entries in the *fstab* file (of course, you have to move the files manually).

Another possible change that you may like to make is regarding the swap partition.

 **Note:** If you have encrypted volumes, you may want to update `/etc/crypttab` as well.